BLG 102E

SSH and Connecting to ev.itu.edu.tr from Windows

You can access the ITU SSH server from your local Windows computer. No additional software installation is required, and it provides an effective and reliable environment for debugging.

To access your folders on the server and upload your code for debugging, you can mount your home directory directly. To do this, your computer must be connected to the ITU network, either through ITU Wi-Fi services (such as ITUNET_Misafir or eduroam) or via ITUVPN.

How to Mount Your Home Directory

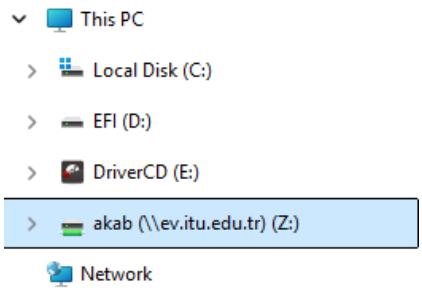
To mount your home directory, use a PowerShell terminal as shown in the figure below:

```
PS C:\Users\chess> net use z: \\ev.itu.edu.tr\akab
Enter the user name for 'ev.itu.edu.tr': itu\akab
Enter the password for ev.itu.edu.tr:
The command completed successfully.

PS C:\Users\chess>
```

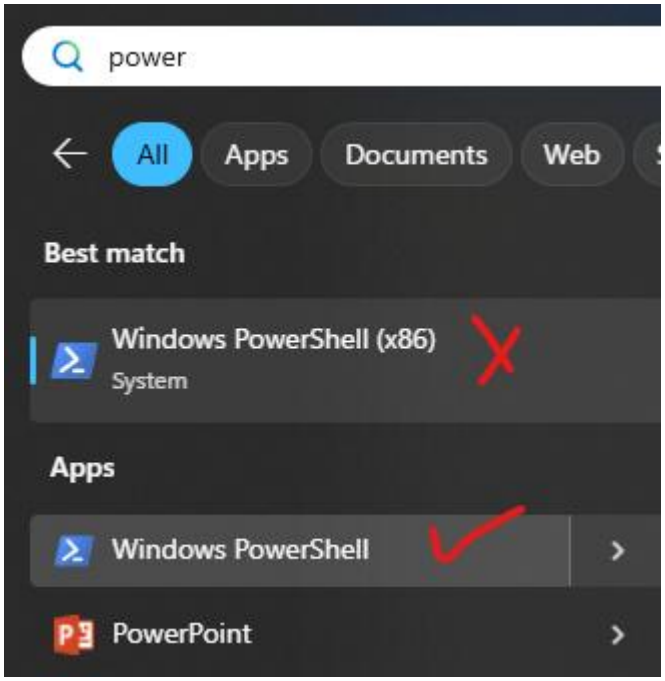
Note that **akab** in this example should be replaced with your own ITU username. Also note that, in the second line, the requested username must include the prefix **itu** before your username.

When the command completes successfully, you will be able to access your home directory as shown below:



Verify SSH Installation

To access a terminal on remote computers in the ITU servers, you can use SSH. To do this, open **PowerShell**. Note that Windows provides two versions of PowerShell; make sure not to use the **x86** version.



Type **ssh** in the terminal. If you see an output similar to the one shown in the image below, your Windows system is already set up to use SSH.

```
PS C:\Users\chess> ssh
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface] [-b bind_address]
           [-c cipher_spec] [-D [bind_address:]port] [-E log_file]
           [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
           [-J destination] [-L address] [-l login_name] [-m mac_spec]
           [-O ctl_cmd] [-o option] [-P tag] [-p port] [-Q query_option]
           [-R address] [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
           destination [command [argument ...]]
PS C:\Users\chess>
```

Otherwise, you need to install or enable OpenSSH on Windows by following the instructions in the link below.

https://learn.microsoft.com/en-us/windows-server/administration/openssh/openssh_install_firstuse?tabs=gui&pivots=windows-11

Accessing ITU Servers Using SSH

In PowerShell, use the **ssh** command to connect to a remote computer.

```
PS C:\Users\chess> ssh ssh.itu.edu.tr -l akab
akab@ssh.itu.edu.tr's password:
Last login: Sun Jul 21 10:37:00 2024 from 160.75.141.124

##################################################################################
##################################################################################
##                                                                              ##
## Kullanicilarimizin Dikkatine!                                                ##
##                                                                              ##
## SSH Sunucumuzun kullanici limitleri asagidadir.                              ##
## Sunucu uzerinde islem yaparken bu limitleri goz onunde bulundurunuz.         ##
##                                                                              ##
## Iyi calismalar.                                                              ##
## ITU BIDB                                                                     ##
## Ag ve Sistem Yonetim Mudurlugu                                               ##
##                                                                              ##
## Tek bir kullanici icin maksimum proses sayisi: 42                            ##
## Tek bir kullanici icin maksimum proses suresi: 300sn                         ##
## Tek bir kullanici icin bellekteki kilitlenen maksimum veri miktari: 40960KB  ##
## Tek bir kullanici icin ayni anda maksimum oturum sayisi: 15                  ##
##                                                                              ##
##################################################################################
##################################################################################

[akab@ssh ~]$ |
```

If you see a screen similar to the one shown above, you have successfully connected to the remote computer. You do not need to be connected to the ITU network to access the remote computer using SSH.

If this is your first time connecting, you will see a prompt similar to the one shown in the image below. Simply type YES to add ITU's SSH key to your list of known hosts.

```
The authenticity of host 'ssh.itu.edu.tr (160.75.25.122)' can't be established.
ED25519 key fingerprint is SHA256:lgpKjm0b53yg1BkUEgY1QM07YztZoBW/FsLVcLW4wxM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ssh.itu.edu.tr' (ED25519) to the list of known hosts.
```

The terminal you access using SSH operates within your home directory, which you mounted in the previous steps.

Alternative:

Alternatively, you can use SSH and SFTP clients (such as MobaXterm) to access your home directory and connect to the SSH server.